AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 1 · STUDENT LESSON

The ML Lifecycle

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Define the task. Fit from training. Freeze before testing.

Distinguish a fitted parameter from a fixed instruction using complete fictional cards.

Keep one habit: Training-only grid → candidate scores → frozen test → twelve-card relay/four probes → independent fit.

MAIN LESSON · SECTION 1

What was chosen from data, and what was written in advance?

> Trace task definition, training-only selection, freezing and a bounded test result.

Your learning goals

I can distinguish **fitting** a rule from applying a fixed rule.

I can trace a small learning lifecycle with separate data.

I can report test **evidence** with its limits.

Recall evidence and limited conclusions

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Creator Defense. Give an example and a mistake that the idea helps you avoid.
- After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.
- A fixed rule matches 3 **label**s out of 4 tested cases. What fraction is correct, and does that prove it works on all future cases?

Your end goal

By the end, I can explain the ml lifecycle using a traced example and a stated limitation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Machine learning is an **evidence** cycle: define a **prediction** problem, prepare examples, fit a model, evaluate on unseen data, and improve one diagnosed cause.
</details>



LEARNING TARGET A team **label**s photos, fits a model and celebrates one correct new photo. List the missing **evaluation** steps and define the intended use before testing.

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

Choose a rule from labelled training examples

> People define the task and rule family; examples determine the selected parameter.

Understand fitting versus a fixed instruction

In Grade 7 you programmed an activity chooser using a rule supplied by a person. This week you will choose a rule by comparing its **prediction**s with **label**led examples.

This small supervised learning task makes the learning **process** visible: people define the task and possible rules; **training** data determines which candidate fits best.

Imagine fictional practice cards with a measured width x and a category **label**, 0 or 1. Width uses invented units. We want to predict the category of another card from its width.

The **label**s are provided **observation**s in this exercise, not values generated by our **prediction** code. These invented cards demonstrate a method; they provide no **evidence** about real objects or students.

Read training cards before final-test labels

Training card	Width in invented units	Supplied category
T1	1	0
T2	3	0
T3	5	1
T4	7	1

Fix candidates and tie rule before testing

Before **fitting**, set aside the final two cards. They are visible here so you can follow the complete example, but do not use them to choose a rule. Our candidates predict 1 when width is at least 2, 4 or 6, and 0 otherwise.

Choose the candidate with the fewest **training** mistakes. If candidates tie, choose the smaller threshold; fix this rule before testing.

▼ Reference · Complete teaching record including held-out cards

Purpose	Width x	Observed label
Training	1	0
Training	3	0
Training	5	1
Training	7	1
Held out for later testing	2	0
Held out for later testing	6	1

▼ Earlier wording, alternatives and picture transcripts

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use the ml lifecycle to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or **evaluation** decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Score candidates on training data and freeze one

> Do not use final-test labels to improve this choice.

Complete training-only candidate loop

```
train = [(1, 0), (3, 0), (5, 1), (7, 1)]
# Predict 1 when x is at least the threshold; otherwise 0.
def predict(x, threshold):
    return int(x >= threshold)

for threshold in [2, 4, 6]:
    errors = 0
    for x, label in train:
        if predict(x, threshold) != label:
            errors = errors + 1
    print(threshold, errors)
```

Inspect all training predictions

Candidate threshold	Training predictions for widths 1,3,5,7	Mistakes
2	0,1,1,1	1 at width 3
4	0,0,1,1	0
6	0,0,0,1	1 at width 5

Explain error counts and record the choice

int(True) is 1 and int(False) is 0. For threshold 2, the **training prediction**s are 0, 1, 1, 1: the width 3 card is wrong. For threshold 4, they are 0, 0, 1, 1: none is wrong.

For threshold 6, they are 0, 0, 0, 1: the width 5 card is wrong. The printed rows are 2 1, then 4 0, then 6 1. Select threshold 4 using this **training evidence** and record it.

The loop reports errors; the learner makes and records the selection.

Selection record: choose threshold 4 because it has the fewest training mistakes, then write the choice before reading the held-out results. Smaller threshold breaks a training-error tie; test labels must not break it. The original code below prints candidate errors; it does not automatically select one. A complete automated selection/test version is supplied later in the investigation.

This worksheet is a visible teaching fixture, not a private final evaluation.

Trace the lifecycle stages

Keep fitting and testing separate

STAGE	INFORMATION USED	RECORD
Define and split	Task and candidate rules	Training and test cards
Fit and freeze	Training labels only	Threshold 4 selected
Test and report	Two held out cards	2 correct out of 2

Read down the stages. Test results may inform later development, but they must not silently change the rule being tested.

Explain the deciding check

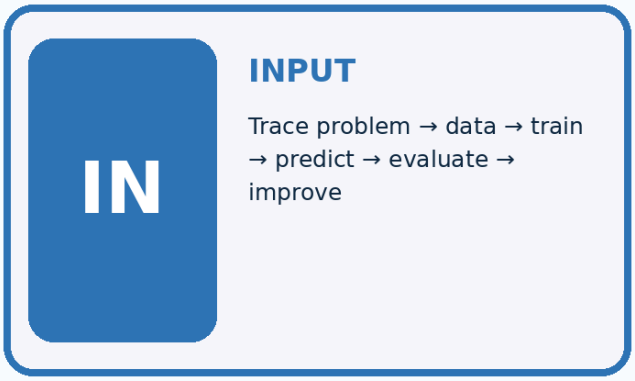
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

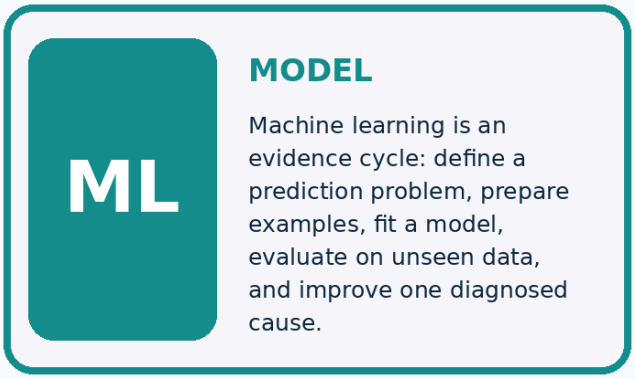
▼ **Earlier wording, alternatives and picture transcripts**

ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and **evaluation** proof separately.







QUESTION Trace problem → data → train → predict → evaluate → improve, and identify where human judgment enters.

MODEL IDEA **Machine learning** is an **evidence** cycle: define a **prediction** problem, prepare examples, fit a model, evaluate on unseen data, and improve one diagnosed cause.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Goal: classify a leaf as healthy or spotted for a classroom plant helper. Prepare **label**ed leaf examples and reserve unseen test examples.

Draw a fresh data → model → **prediction** → **evaluation** trace:

A closer explanation

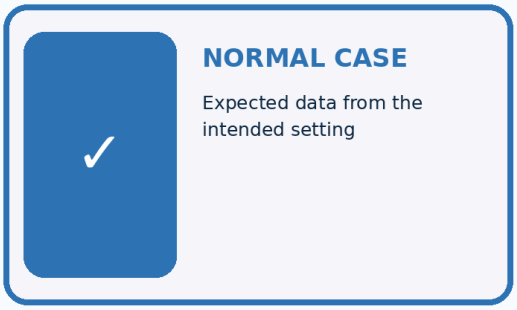
Connect the picture and the explanation

MAIN LESSON · SECTION 4

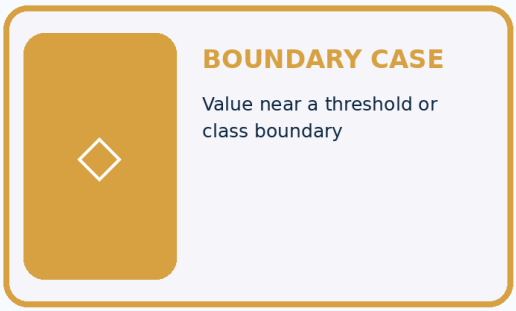
Compare normal, boundary, missing and changed-reference evidence

> Each probe has a different question and evidence requirement.

Four evaluation conditions



NORMAL Expected data and expected behavior.

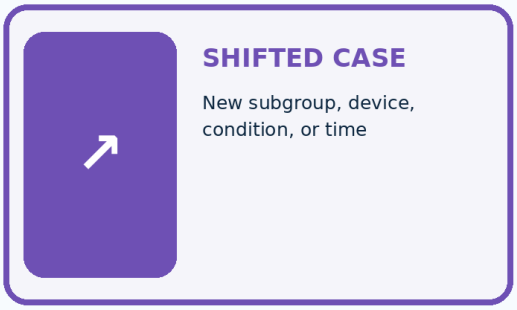


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous **label**, or uncertain **prediction**.



MISSING Necessary field, **label**, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: The fresh width-6 leaf is actually healthy.

▼ **Check after predicting**

Prediction: The frozen model is correct on 1/2 test leaves.

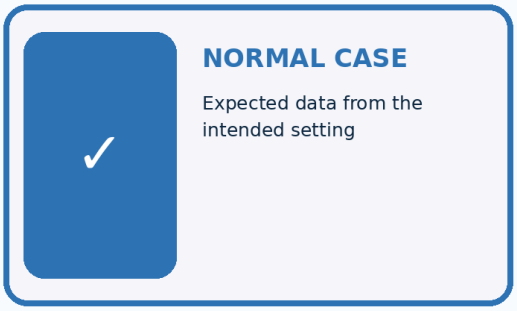
Reason: The supplied reference changed; moving the cutoff after seeing this test would reuse final **evidence** for development.

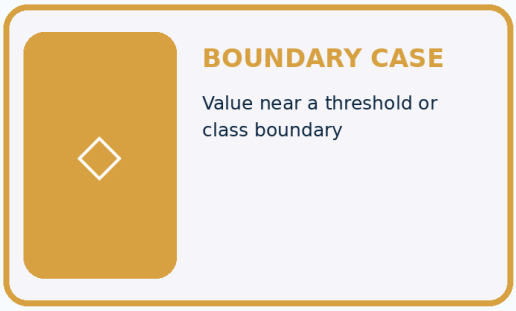
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

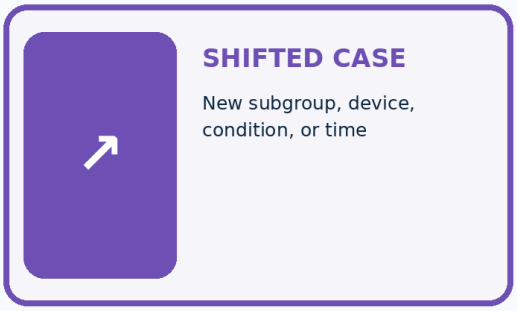
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous **label**, or uncertain **prediction**.

MISSING Necessary field, **label**, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Test a frozen rule, then preserve the changed-reference error

> The main and pictured candidate sets are labelled separately.

Read the complete pictured training and test fixture

Follow one complete plant-classifier lifecycle without tuning on the final test.

Toy **training** leaves have widths 1 healthy, 3 healthy, 5 spotted, 7 spotted. Candidates predict spotted at width>=2 or >=4.

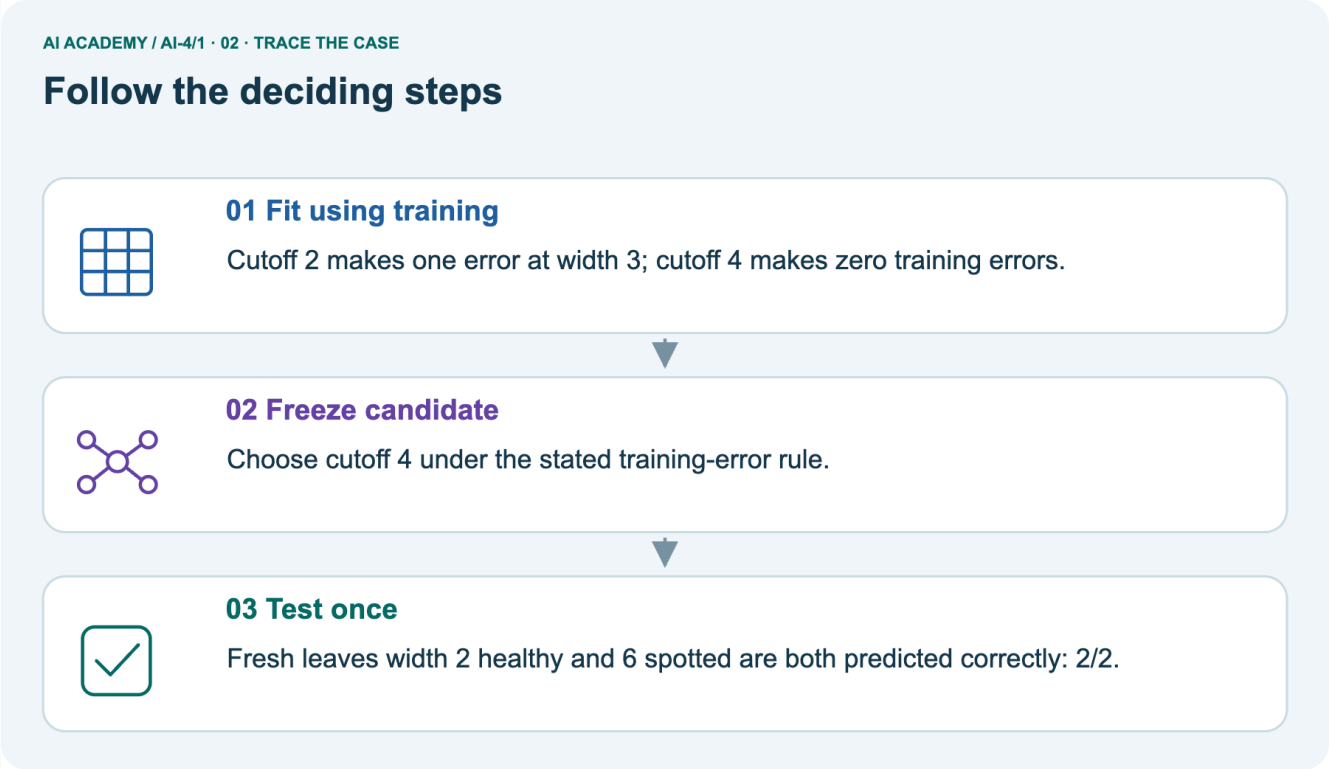
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The main numeric example considers thresholds 2,4,6 and chooses 4. The separate pictured plant-label fixture considers only 2 and 4 and also chooses 4; its healthy/spotted words are fictional classroom labels, not botanical diagnosis. With frozen 4, held-out widths 2 and 6 get

predictions 0 and 1. Changed reference: width 6 is now labelled healthy/0, so the frozen prediction is wrong and the score is 1/2.

Preserve that result. The reference change is supplied, not evidence that width causes a health condition or that real distribution shift was measured.

Trace fitting, freezing and evaluation



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Fit using training	Cutoff 2 makes one error at width 3; cutoff 4 makes zero training errors.
2. Freeze candidate	Choose cutoff 4 under the stated training-error rule.
3. Test once	Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

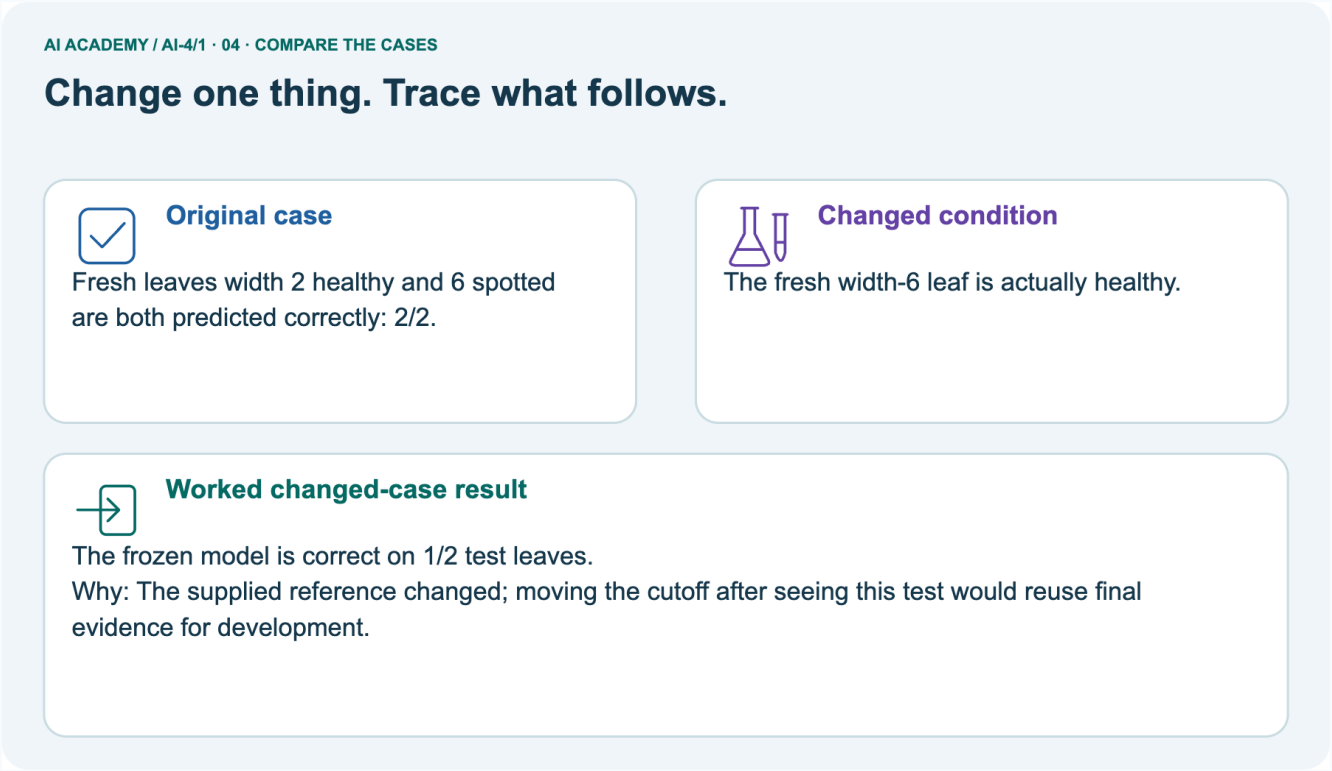
Read the picture in order. Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict changing the width 6 reference label

- Change one thing: The fresh width-6 leaf is actually healthy. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State the two-card and real-use limits

- Two invented test leaves cannot establish useful real-world plant classification.
- Read the **label**led picture: Should the team silently retune on the changed test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

▼ Reference · Main numeric held-out result

Freeze means do not change the chosen threshold during this **evaluation**. With threshold 4, widths 2 and 6 receive **prediction**s 0 and 1. Compare them with the held out **label**s: both match. Report 2 correct out of 2 held out cards.

Training results and test results answer different questions: did the rule fit the examples used to select it, and how did that fixed rule perform on these separate examples?

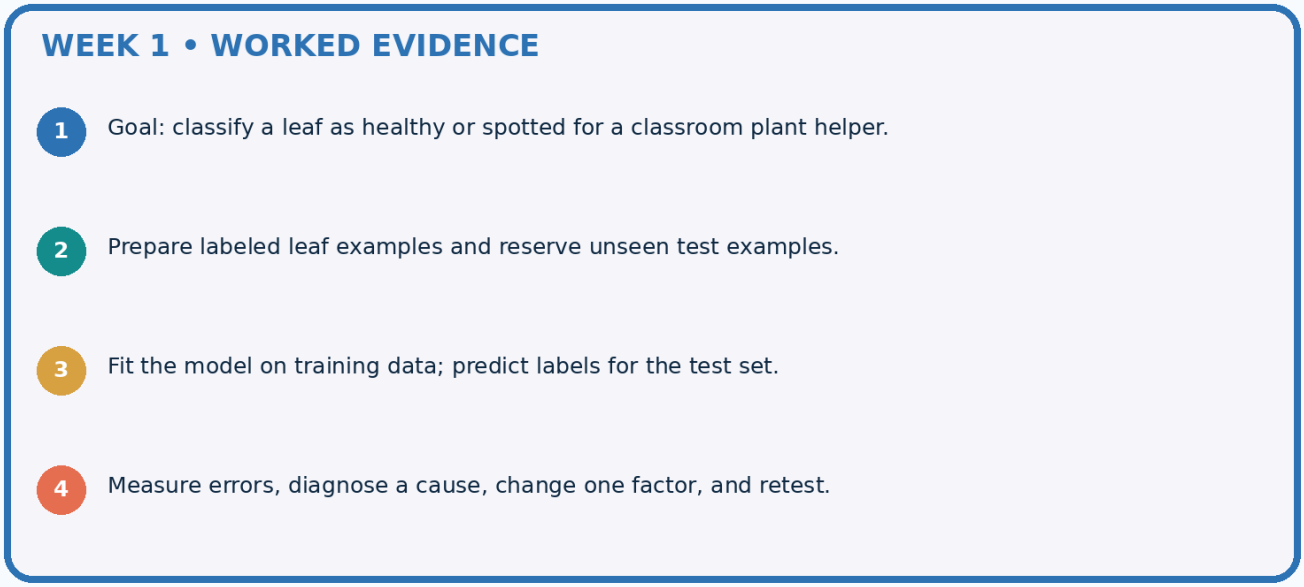
> **Keep threshold 4 fixed. An additional card has width 4 and observed label 0. Predict its category, decide whether it is correct, and update the held out count from 2 of 2.**

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.



STEP 1 Goal: classify a leaf as healthy or spotted for a classroom plant helper.

STEP 2 Prepare **label**ed leaf examples and reserve unseen test examples.

STEP 3 Fit the model on **training** data; predict **label**s for the test set.

STEP 4 Measure errors, diagnose a cause, change one factor, and retest.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

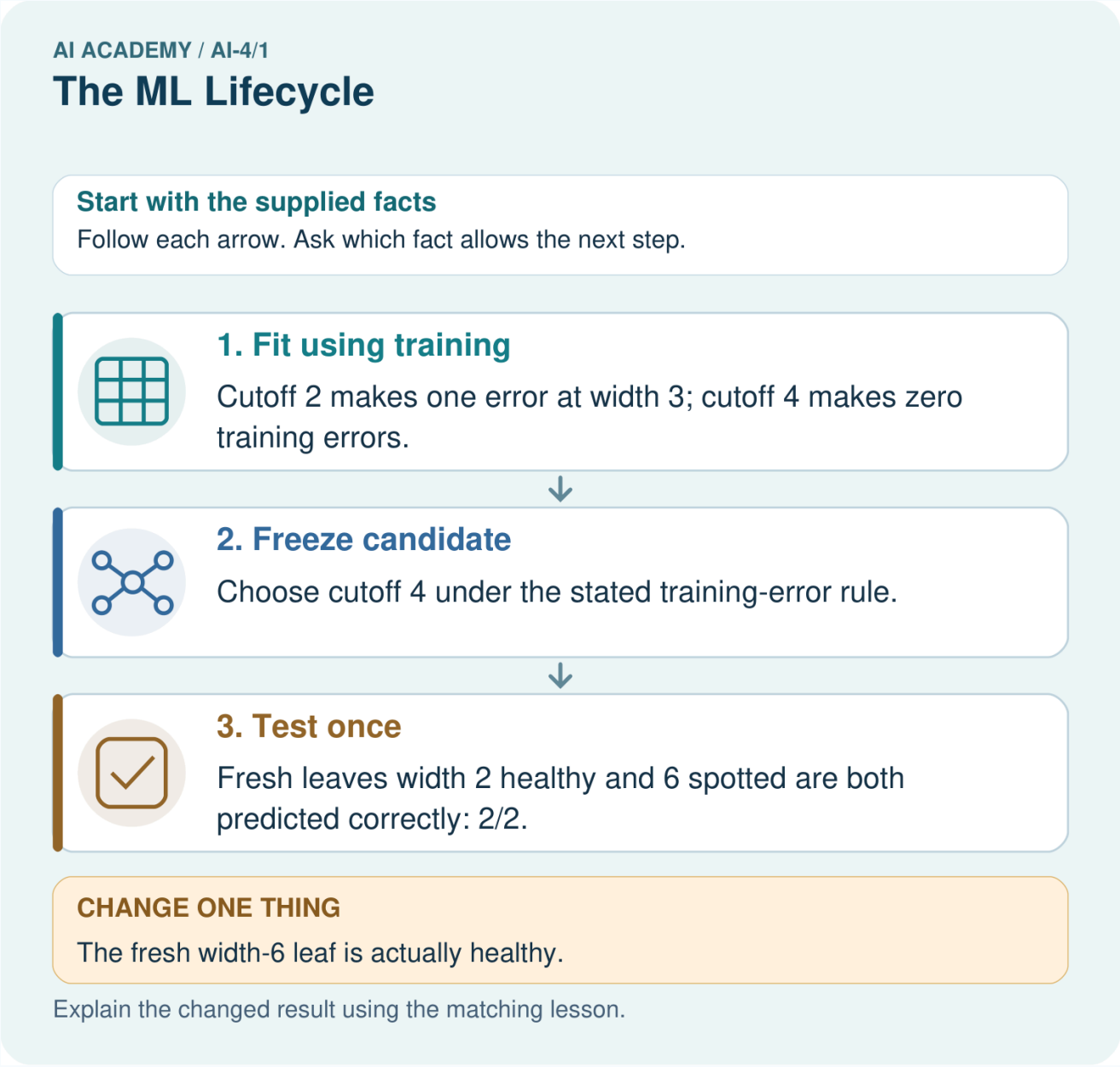
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: The ML Lifecycle

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find the choice that used test information

> A later revision cannot reuse an exposed result as untouched final evidence.

Read the first-error diagnosis table

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

State the small rule-family limitation

- This threshold model learned only a choice among three candidate values. People supplied the **feature**, **label**s, candidate family and tie rule.

  Learning does not imply understanding, and perfect **training** performance does not guarantee good **prediction**s. Two correct test cards are too few to establish reliable performance for a broader population.

  Even a larger score would depend on how examples were collected and what use was intended.

Your report says the fixed model matched 2 of 2 held out cards. Write one supported claim, one claim this does not support, and one sensible next step.

Challenge one-correct-prediction proof

- Diagnose one cause and name the cleanest next experiment:

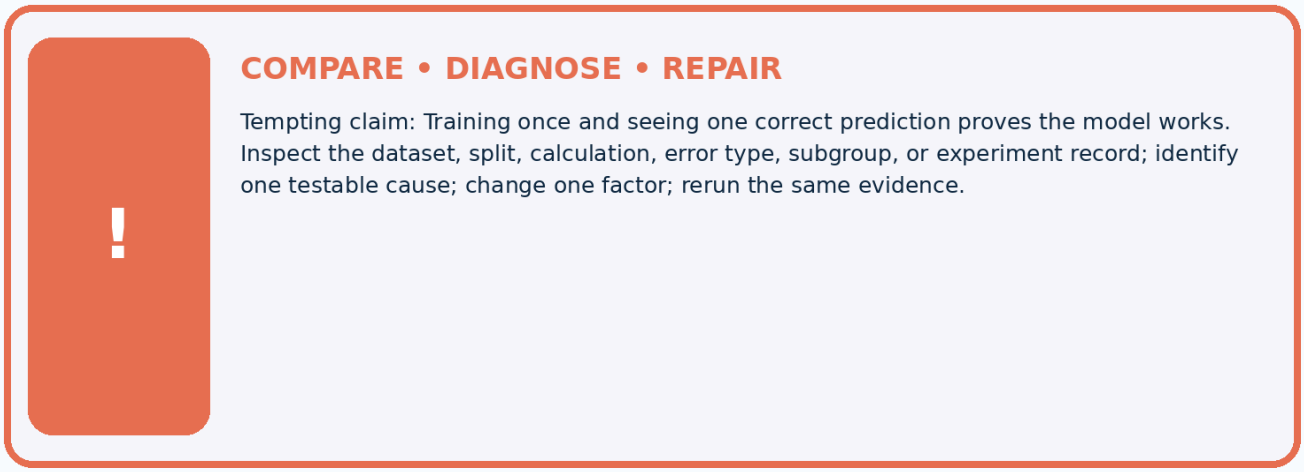
CLAIM TO CHALLENGE **Training** once and seeing one correct **prediction** proves the model works.

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



A closer explanation

MAIN LESSON · SECTION 8

Practise threshold 6 without changing the training labels

> Record every prediction and error before comparing candidates.

Complete supported W2

Use workbook W2 for the connected example “The **machine learning** lifecycle”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Audit who chose the task and rule family

- Use the worked case for The **machine learning** lifecycle. Proposed explanation: “The model wrote its own task”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured frozen-model change

- Should the team silently retune on the changed test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve independent fitting

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Additional development probe · Width4error and retuning

A learner sees the width 4 test error, changes the threshold to 5, and reports the same cards as a fresh final test. The test **label**s have now influenced the model. Keep the original 2 of 3 result for the original rule.

Record the change as development, then reserve genuinely unused examples for a later **evaluation**. Do not delete the difficult card or rename reused cards as new **evidence**.

A lifecycle continues after **fitting**: define the intended use, prepare and separate data, fit, evaluate, review the limits, and monitor any approved use. **Evaluation** can send us back to development.

That return needs an honest record and a new final **evaluation**; it does not erase the earlier error.

▼ Reference · Experiment-record fields

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + **feature**s + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

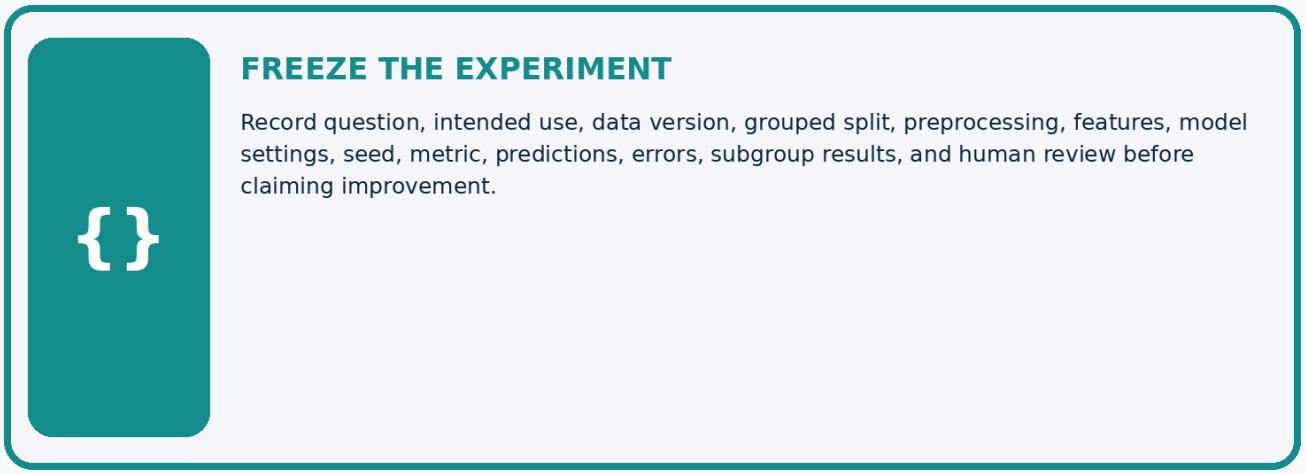
Write the minimum record needed to reproduce today's result:

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and **evaluation** rule.



Experiment record

A closer explanation

Practise with less help: Trace one candidate

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete twelve-card relay and four evidence probes

> No action cards, broken workflows or probe labels are left for the teacher to invent.

Twelve shuffled action cards

Action card	Complete fictional lifecycle action
F	Count candidate mistakes on training cards only.
A	Define the intended classroom task and observed label meaning.
J	Compare frozen test predictions with their supplied labels.
C	Collect the supplied fictional cards and their provenance.
E	Predeclare candidate thresholds and smaller-threshold tie rule.
L	Plan revision using development evidence and newly reserved final cases.
H	Freeze the chosen parameter for this evaluation.
B	Specify width feature, invented units and prediction-time availability.
K	Keep raw outcomes, error count and bounded conclusion.
D	Declare training/test card roles before fitting.
I	Predict the held-out cards without fitting on them.
G	Select and record the threshold from training scores.

All four declared probe inputs

Separate complete probe under frozen threshold 4	Expected evidence
Normal original test: widths 2 label 0 and 6 label 1	Predictions 0,1; 2/2 correct on these cards
Additional boundary DB1: width 4, reference 0	Prediction 1, wrong on this separate development probe

Separate complete probe under frozen threshold 4	Expected evidence
Missing feature DM1: width unavailable, reference 0	No supported numeric prediction; request feature, do not substitute 0
Changed-reference DS1: width 6, reference 0	Prediction 1, wrong; original reference change must remain visible

Arrange, diagnose, predict and record

1. Lifecycle Relay: arrange all twelve supplied action cards and justify the dependencies. One valid order is A–L; some preparation steps may move if training/test roles and the selection rule are fixed before fitting. Repair workflow X that fits on all six cards then calls them independent, and workflow Y that changes threshold after a test error but overwrites the original score.

   Use the four complete probes separately, preserving numeric prediction, missing-feature status and reference scope. Predict training scores/choice before the optional complete Python run. That program assumes the supplied complete numeric training rows; it does not handle DM1 as a raw input validator. Toy code execution is allowed evidence of this computation, not evidence of real plant usefulness.

   Keep independent candidates 3/5/7 and their new cards withheld.

Complete optional fitting/evaluation program

```
def predict(width, threshold):
    return int(width >= threshold)

def fit_threshold(training_rows, candidates):
    if not training_rows or not candidates:
        raise ValueError('training rows and candidates required')
    scored = []
    for threshold in candidates:
        mistakes = sum(predict(width, threshold) != label
                       for width, label in training_rows)
        scored.append((mistakes, threshold))
    # Tuple order implements fewest errors, then the smaller threshold.
```

```
    return min(scored)[1], scored

train = [(1, 0), (3, 0), (5, 1), (7, 1)]
threshold, scored = fit_threshold(train, [2, 4, 6])
print('training', scored)
print('frozen threshold', threshold)
# These rows are used only after selection in this complete classroom example.
test = [(2, 0), (6, 1)]
predictions = [predict(width, threshold) for width, label in test]
correct = sum(pred == label for pred, (width, label) in zip(predictions, test))
print('test predictions', predictions)
print('test correct', correct, 'of', len(test))
```

Keep the experiment ledger

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

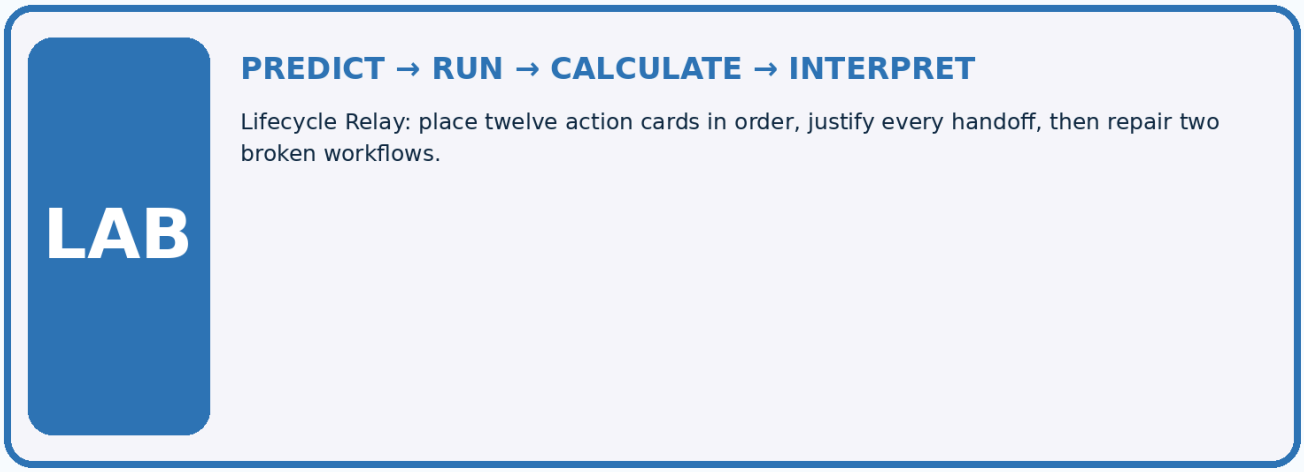
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Lifecycle Relay: place twelve action cards in order, justify every handoff, then repair two broken workflows.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: fit and evaluate the new card set

> Keep fitting, test prediction and interpretation as separate stages.

Independent cards and scope

New **training** cards are (width 2, **label** 0), (4, 0), (6, 1), (8, 1). Candidates are 3, 5 and 7. Predict 1 when width is at least the threshold.

Count **training** mistakes, select the fewest with smaller threshold breaking ties, and record the choice before testing. Held out cards are (3, 0), (7, 1), (5, 0).

Give each test **prediction**, the correct count and a limited conclusion.

Use this task's new training cards and candidate thresholds. Record the training mistake counts and selected threshold before checking the three held-out outcomes. Preserve the frozen choice even if a final card fails; a new revision would need a separate evidence record and fresh evaluation.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Independent case: Fit and evaluate independently

Week 1 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A team selects a model before deciding what outcome to predict or where the data comes from. Name two missing lifecycle decisions and their effect.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result
</details>

Write or draw your reasoning and evidence.

Connect to your project

Write the Responsible Plant Classifier problem statement, intended user, **prediction** target, success measure, and human-review point.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

A team **label**s photos, fits a model and celebrates one correct new photo. List the missing **evaluation** steps and define the intended use before testing.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Keep the first fitted decision and its evidence

> The classifier project begins with a bounded problem and data roles.

This week’s contribution

For the Responsible Plant Classifier, record the intended task, feature/label definitions, source/split versions, fixed candidate family, tie rule and frozen parameter. Preserve every prediction and error. These width cards teach parameter fitting; they do not validate a real plant tool. Plan new data and human review before any broader usefulness claim, keeping already exposed probes in development.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Keep checkpoint, limits and human decision

THIS WEEK'S CHECKPOINT Write the Responsible Plant Classifier problem statement, intended user, **prediction** target, success measure, and human-review point.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: name the fitted choice and what the test establishes

> A rule selected from data still has a people-defined task and evidence boundary.

Explain without notes

- Close your notes. Explain this goal in your own words: Distinguish **fitting** a rule from applying a fixed rule

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
machine learning	fit a model from examples	
training	adjust model parameters using examples	
inference	apply the fitted model to new input	
evaluation	measure behavior on unseen examples	

Closer-explanation terms

Word	Meaning
Feature	An input value used to make a prediction
Label	An observed target value supplied for an example
Training	Using examples to select or adjust a model

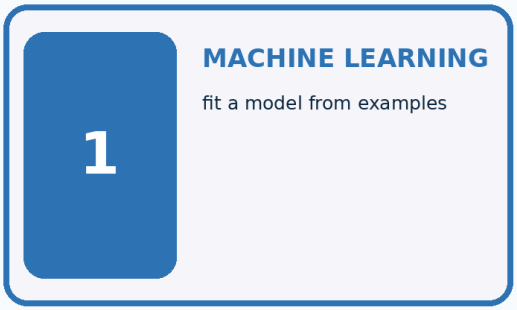
Word	Meaning
Held out data	Examples reserved from fitting and model choices for evaluation

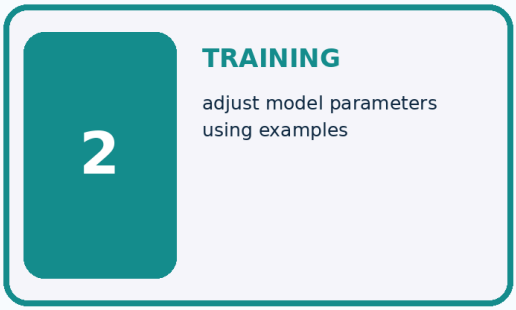
▼ Earlier wording, alternatives and picture transcripts

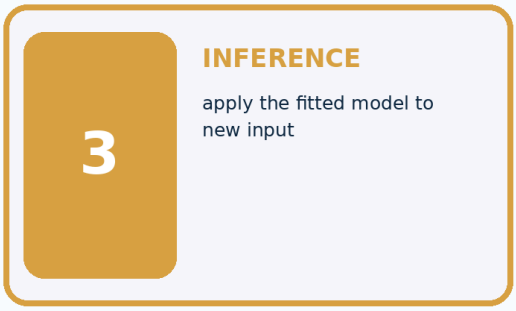
VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR **Training** once and seeing one correct **prediction** proves the model works.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **machine learning**, **training**, **inference**, **evaluation**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE **Training** once and seeing one correct **prediction** proves the model works.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

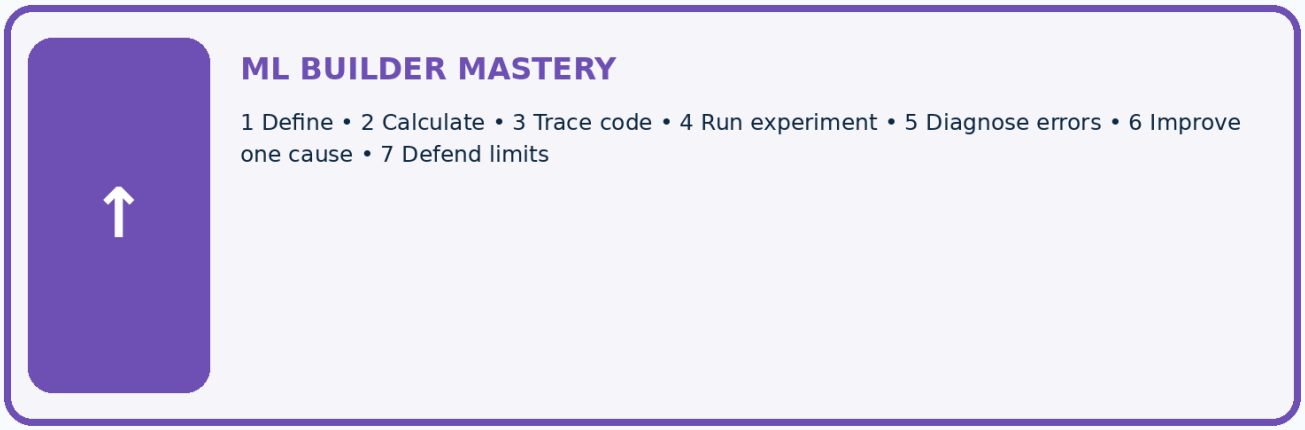
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep Fit and evaluate the new card set independently for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, **feature**/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

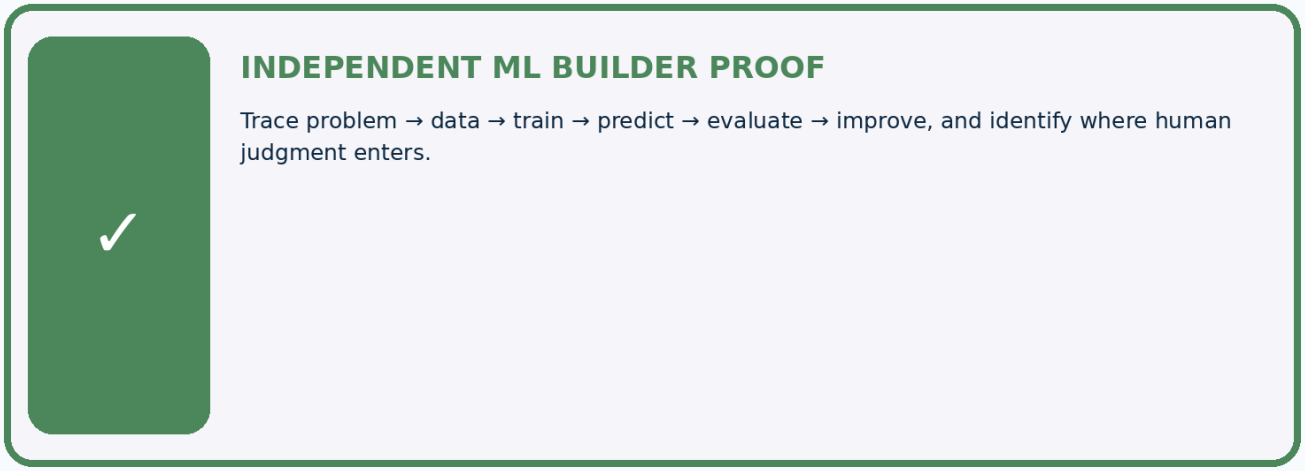
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Trace problem → data → train → predict → evaluate → improve, and identify where human judgment enters.

NEXT BRIDGE Carry the experiment **evidence** into: **Feature**s, **Label**s, Targets.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

A team **label**s leaf photos, trains a healthy/spotted classifier and shows one correct **prediction**.

Order the lifecycle and identify missing **evidence**.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Define use and **label**s, collect data, separate **evaluation** data, train, predict, evaluate and improve. One correct result is insufficient. Test unseen varied cases and retain human responsibility.

Evidence or stage	Value or result
Define	Intended use and labels
Prepare	Collect and separate evaluation data
Train	Fit on training data
Evaluate	Inspect unseen results

Evidence or stage	Value or result
Revise	Change a cause and retest

▼ Earlier wording, alternatives and picture transcripts

Week 1 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Evidence drives the ML lifecycle

A team **label**s leaf photos, trains a healthy/spotted classifier and shows one correct **prediction**.

Order the lifecycle and identify missing **evidence**.

Define use and **label**s, collect data, separate **evaluation** data, train, predict, evaluate and improve. One correct result is insufficient. Test unseen varied cases and retain human responsibility.

▼ Reference · Earlier route checklist

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Distinguish fitting a rule from applying a fixed rule	Explain the deciding step, show a trace or test, and state one limit.
Trace a small learning lifecycle with separate data	Explain the deciding step, show a trace or test, and state one limit.
Report test evidence with its limits	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Follow one complete plant-classifier lifecycle without tuning on the final test.

▼ Extra practice · Predict, check and repair

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Follow one complete plant-classifier lifecycle without tuning on the final test.
- Which **label**led step matches this detail? Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2.
- 1. Test once
- 2. Freeze candidate
- 3. Fit using **training**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Test once Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. Read the picture in order.

Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The fresh width-6 leaf is actually healthy. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. The frozen model is correct on 1/2 test leaves.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: The frozen model is correct on 1/2 test leaves. The supplied reference changed; moving the cutoff after seeing this test would reuse final **evidence** for development.

Repair and keep the record

Review Creator Defense (ai-3/36). Goal: Explain the **mechanism** and terminology of generative systems defense. Short practice: Which numbers would you display to make the defense honest?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, **label** your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

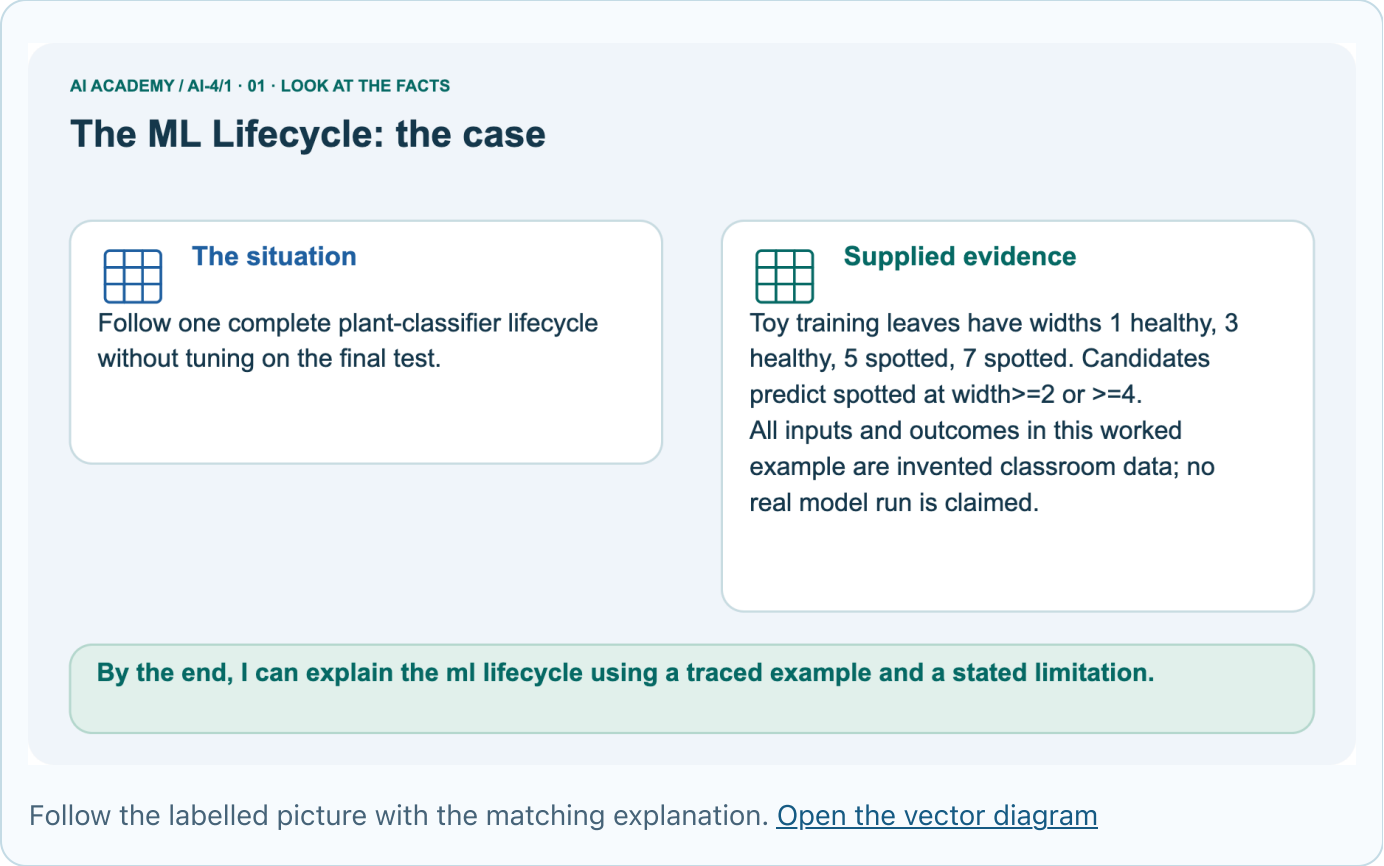
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. The ML Lifecycle: the case



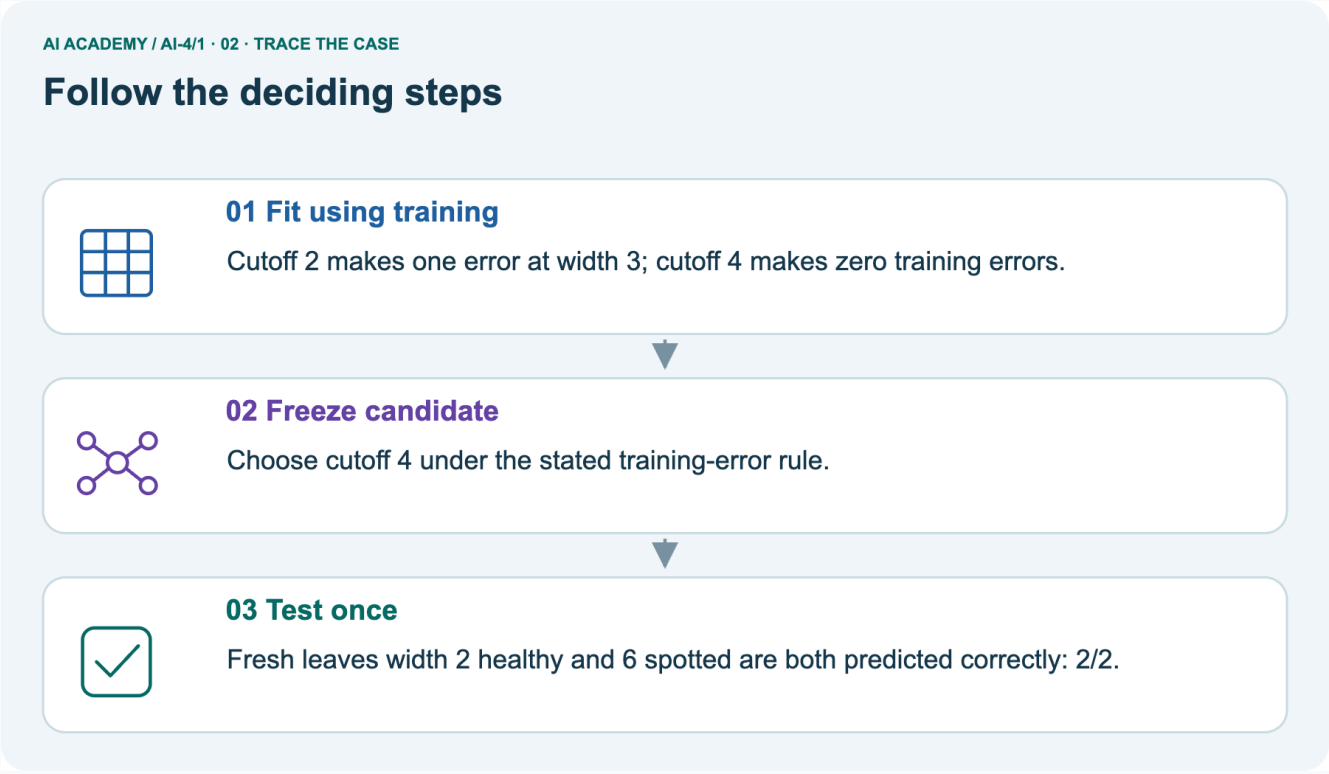
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow one complete plant-classifier lifecycle without tuning on the final test. Supplied facts: Toy **training** leaves have widths 1 healthy, 3 healthy, 5 spotted, 7 spotted. Candidates predict spotted at width>=2 or >=4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

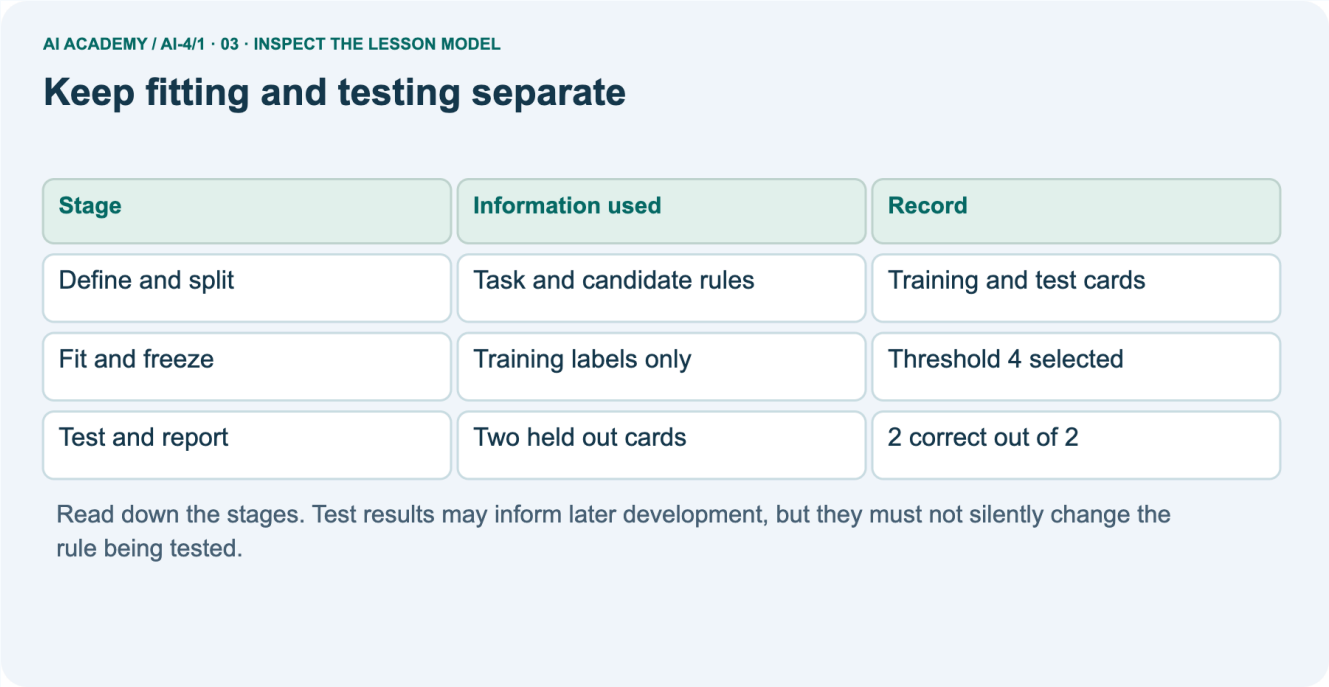


Follow the labelled picture with the matching explanation. Open the vector diagram

Fit using **training**: Cutoff 2 makes one error at width 3; cutoff 4 makes zero **training** errors. → Freeze candidate: Choose cutoff 4 under the stated **training**-error rule. → Test once: Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2.

Point to each numbered stage and explain which recorded **input** it uses.

03. Keep fitting and testing separate

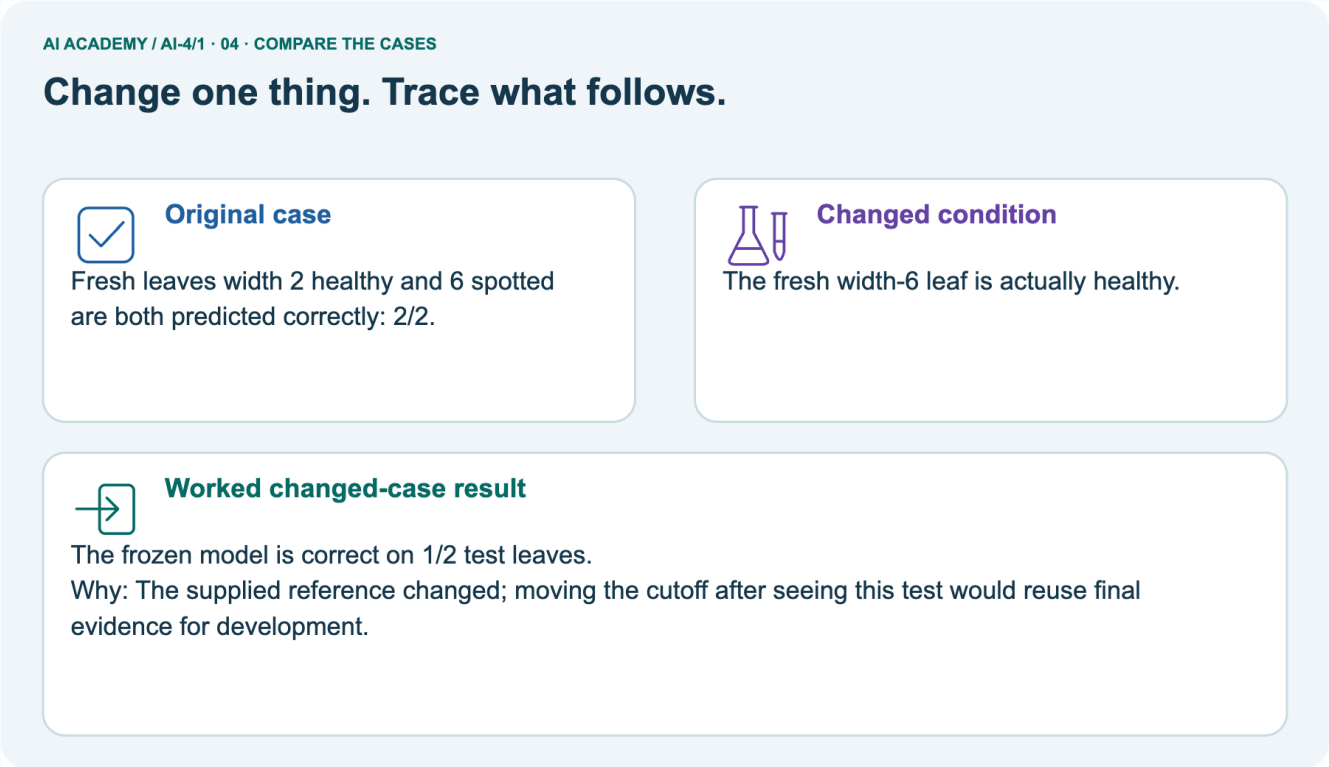


Follow the labelled picture with the matching explanation. Open the vector diagram

Keep **fitting** and testing separate. Define and split | Task and candidate rules | **Training** and test cards. Fit and freeze | **Training label**s only | Threshold 4 selected. Test and report | Two held out cards | 2 correct out of 2

Read the **label**s and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



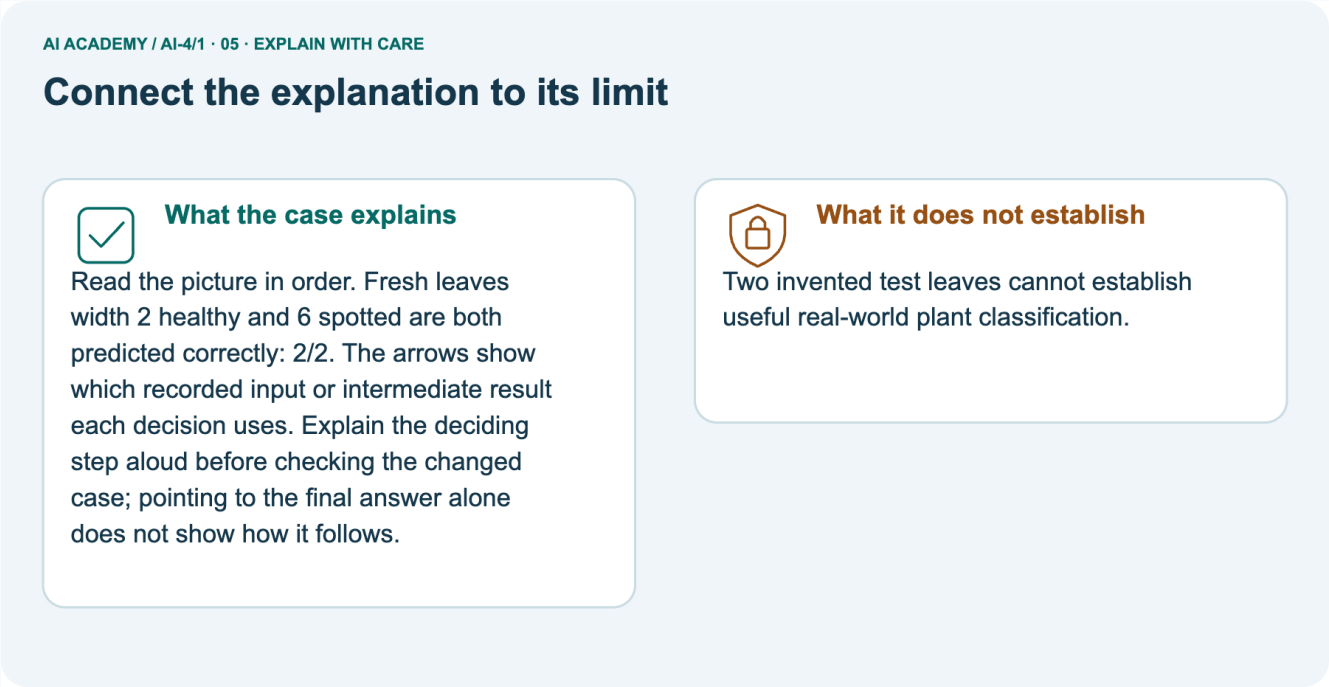
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. Changed condition: The fresh width-6 leaf is actually healthy. Expected result: The frozen model is correct on 1/2 test leaves.

Reason: The supplied reference changed; moving the cutoff after seeing this test would reuse final **evidence** for development.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



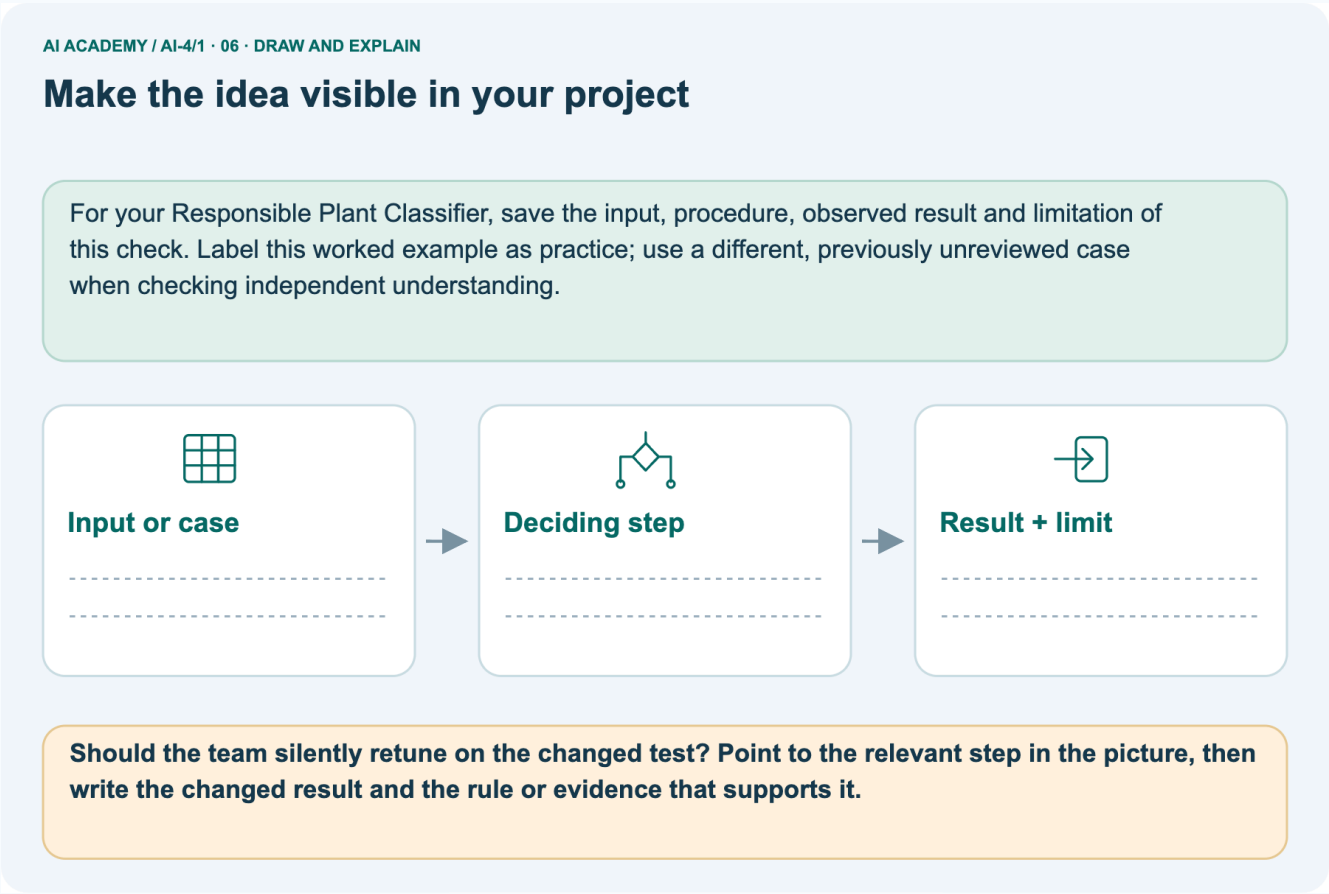
Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Two invented test leaves cannot establish useful real-world plant classification.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Should the team silently retune on the changed test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. The ML Lifecycle: the case

02. Follow the deciding steps

03. Keep fitting and testing separate

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.